

- Q.23 Explain homogeneous and heterogeneous systems.
- Q.24 Explain Boyle's Law
- Q.25 Explain Isothermal process.
- Q.26 Explain Carnot engine with neat sketch.
- Q.27 Compare water tube and fire tube boilers.
- Q.28 Explain Boiler Mounting with Example.
- Q.29 Explain Boiler Accessories in brief.
- Q.30 Explain single stage Compressor with neat sketch.
- Q.31 Write advantages of multistage Compressor.
- Q.32 Explain Axial flow Compressor. With diagram.
- Q.33 Write advantages of super heater.
- Q.34 Explain Kelvin-Planck statement of 2nd Law of TD
- Q.35 Explain thermodynamic properties of steam

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain Construction and working of Centrifugal Compressor with neat sketch.
- Q.37 Explain construction and working of Babcock and Wilcox boiler with neat sketch.
- Q.38 Explain constant Pressure process and Isothermal Process in details.

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Roll No.

2nd Sem / Mech. Engg (MSIL)

Subject:- Thermodynamics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The envelop which separates the thermodynamic system from surrounding is known as
- a) Boundary b) State
- c) Universe d) Process
- Q.2 The condition of perfect vacuum and zero pressure occur at
- a) 0°C b) 273°C
- c) 273°K d) -273°C
- Q.3 The S.I unit of characteristic gas constant is
- a) J/kg b) J/K
- c) J/kg K d) kJ/kg
- Q.4 If a gas is heated at Constant volume
- a) Pressure increase
- b) Temperature increases
- c) Both (a) and (b)
- d) Enthalpy Increases

- Q.5 The efficiency of perpetual motion machine of second kind is
- a) 0% b) 50%
- c) 75% d) 100%
- Q.6 Babcock and Wilcox Boiler is a
- a) Water Tube Boiler
- b) Externally fired Boiler
- c) Multi Tubular Boiler
- d) All of the above
- Q.7 The device which increase the temperature of saturated steam without raising its pressure is known as
- a) Superheater b) Economiser
- c) Air preheater d) Duplex Pump
- Q.8 Mollier diagram is a plot of
- a) Temperature and Entropy
- b) Temperature and Enthalpy
- c) Entropy
- d) Enthalpy and Entropy
- Q.9 The capacity of air Compressor is measured in
- a) m^3/s b) m/s
- c) m^2/s d) ms

- Q.10 Vane blower Compressor is a type of
- a) Centrifugal Compressor
- b) Single stage compressor
- c) Reciprocating Compressor
- d) Rotary Compressor

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define Surrounding.
- Q.12 State Avogadro's law.
- Q.13 What is the other name of Constant Volume Process?
- Q.14 Define throttling process.
- Q.15 Define Heat Source.
- Q.16 What is a boiler?
- Q.17 Write the function of fusible plug.
- Q.18 Name any two boiler accessories.
- Q.19 Define dryness fraction.
- Q.20 Write the function of intercooler.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain open system with an example.
- Q.22 Explain irreversible process.